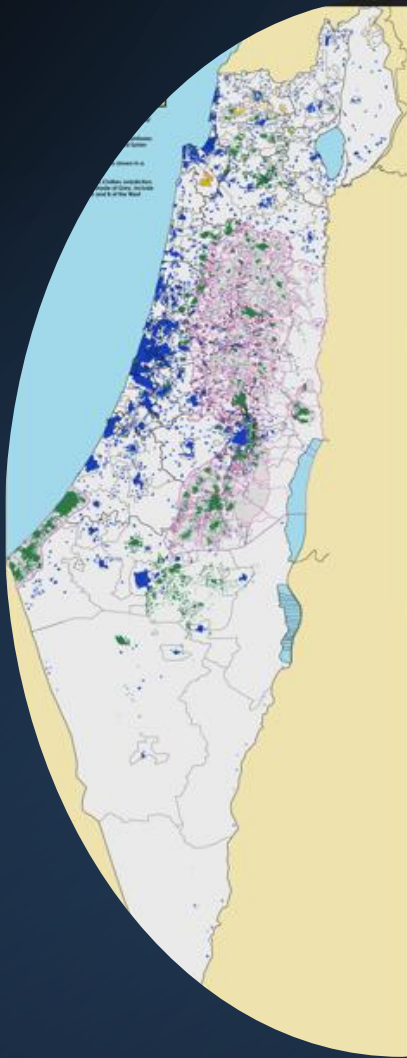




Ethnicity vs. Trust in the Israeli Healthcare System

A Data-Driven Analysis of Healthcare Perception in Israel
Presenters: Ben Vann and Ben Jaitin

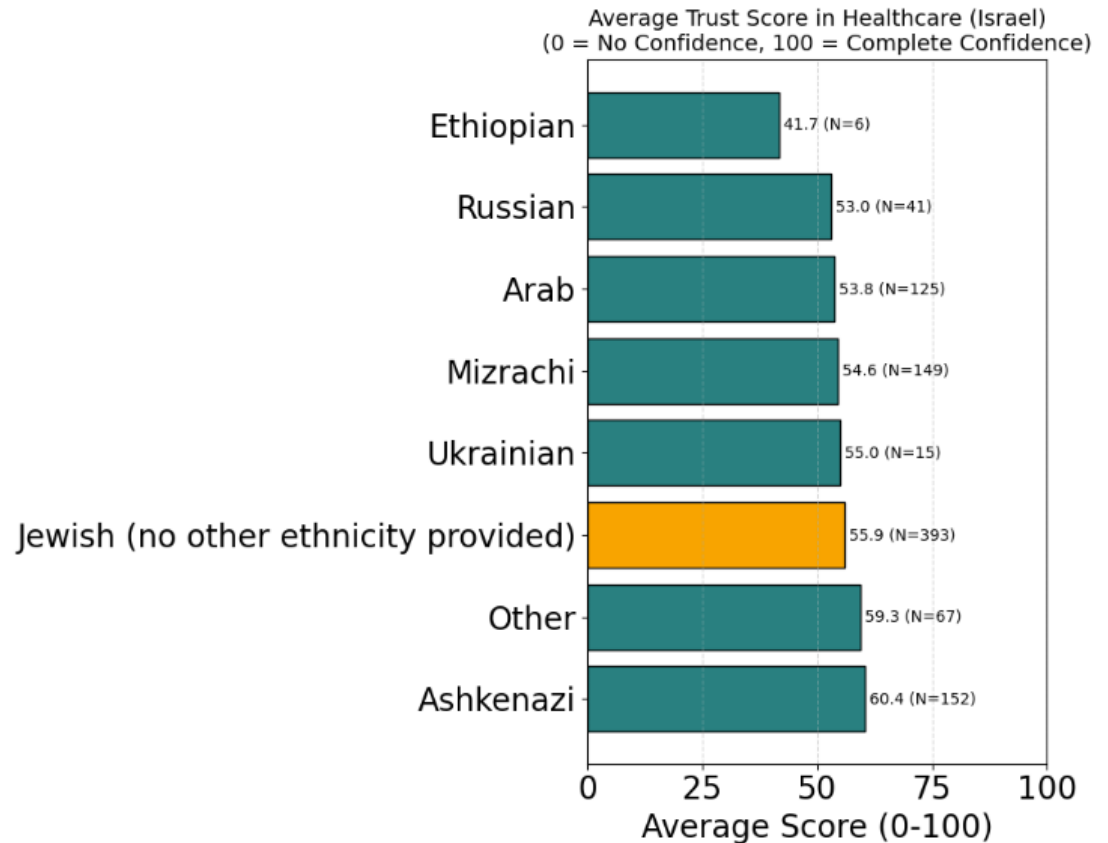
Data Source & Methodology

- Data Source:
 - ISSP 2021 - 'Health and Health Care II' (ZA No. 8000)
 - Dataset filtered for Israeli population (country == 'IL-Israel')
- Ethnic Subsets:
 - Major Groups: Ashkenazi, Mizrachi, Arab, Jewish (General)
 - Minor Groups: Russian, Ukrainian, Ethiopian, Other
- Metric:
 - 'Trust Score' derived from confidence levels (0-100 scale)

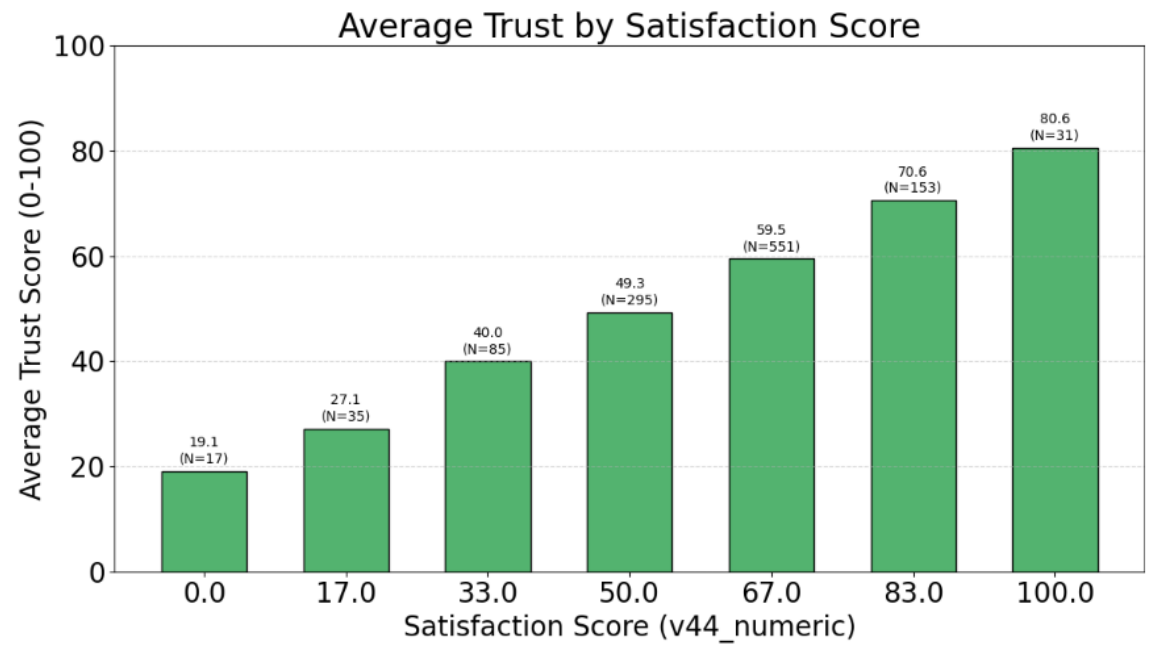
Group	N
Ashkenazi	152
Other	67
Jewish (no other ethnicity provided)	393
Ukrainian	15
Mizrachi	149
Arab	125
Russian	41
Ethiopian	6

Initial Findings (Average Trust Scores)

- Key Insight: Trust varies significantly across ethnic origins



Trust vs. Satisfaction Map



Research Pivot

Research Question:

- What actually drives trust in the healthcare system?
- Is it purely ethnicity, or are there deeper factors?

Pivot:

- Moving beyond simple averages to a predictive regression model.

Key Questions:

- Can we predict trust based on demographics?
- Which factors differ between sectors?

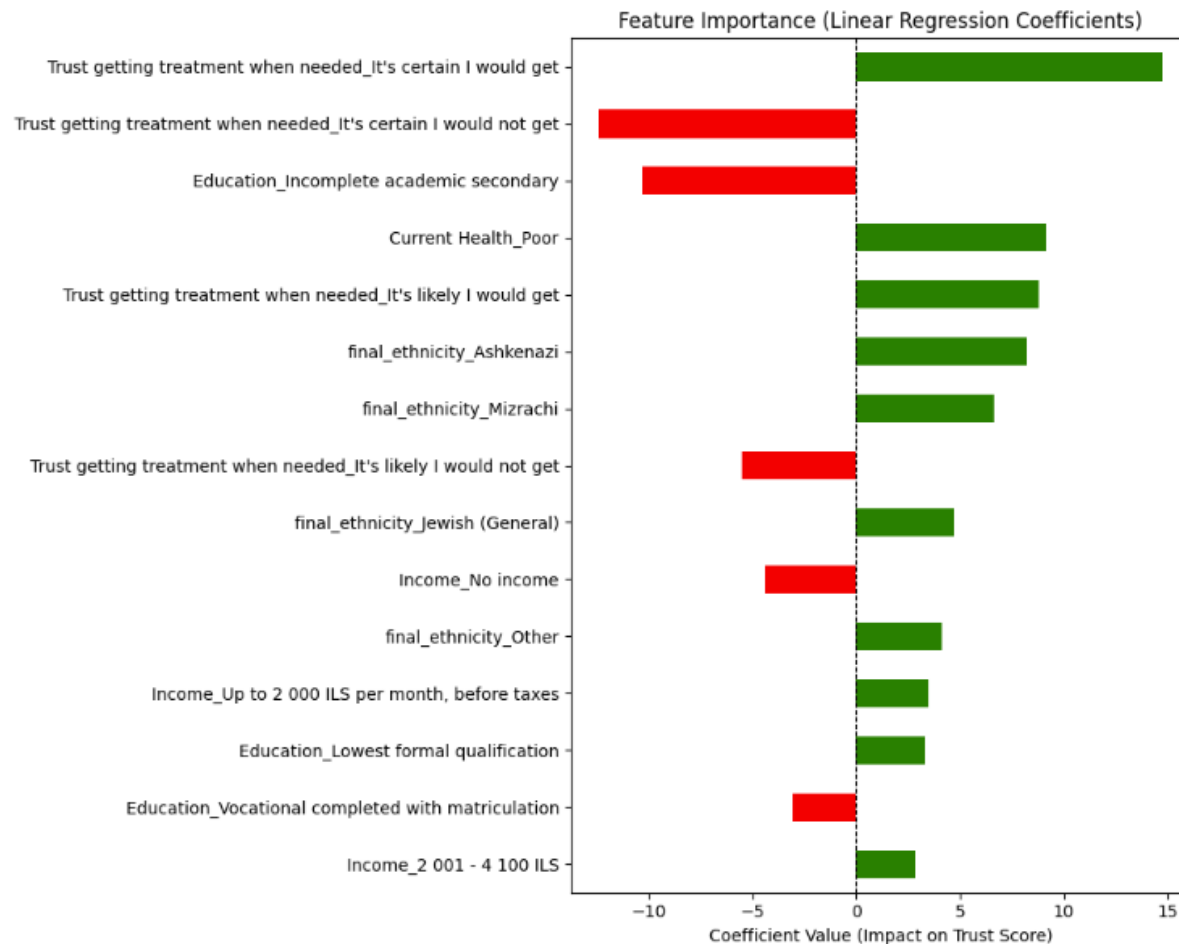
Model 1: Diverse Data Setup

- Objective: Build a model to predict Trust Score.
- Features Selected:
 - Health Behaviors: Smoking habits
 - Perception: Trust in getting treatment
 - Demographics: Current Health, Sex, Education, Income, Ethnicity

```
model_features = ['v43', 'v51', 'v47_fixed', 'sex', 'il_iscd', 'il_rinc', 'final_ethnicity']
existing_features = [col for col in model_features if col in df_isr.columns]

df_model = df_isr[existing_features + ['Trust_Score']].copy()
```

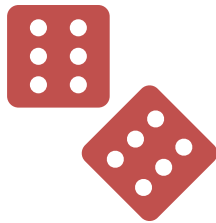
Feature Importance



Feature Importance Conclusions

- Result:
 - The model achieved a low R-squared ≈ 0.174
- Conclusion:
 - Demographics and basic health habits alone explain very little of the variance.
 - The 'Trust' variable remains elusive using standard metrics.

Pivot to Current Health



Hypothesis:

Since 'Self-Reported Health' had a high impact in Model 1, does physical health directly predict trust?



Proposal:

Narrow the focus to strictly health-related predictors.

Current Health Model Setup

- Input: Objective physical health metrics.
- Target: Trust Score.
- Goal: Determine if sick people trust the system less than healthy people.

```
#The proper way to calc BMI
df_isr['BMI'] = df_isr['weight_kg'] / ((df_isr['height_cm'] / 100) ** 2)

#Expanding our parameters for the Linear Regression
features_to_check = ['v43', 'v51', 'v47_fixed', 'sex', 'il_iscd', 'il_rinc', 'final_ethnicity',
                    'age_clean', 'height_cm', 'weight_kg', 'BMI']

existing_features = [col for col in features_to_check if col in df_isr.columns]

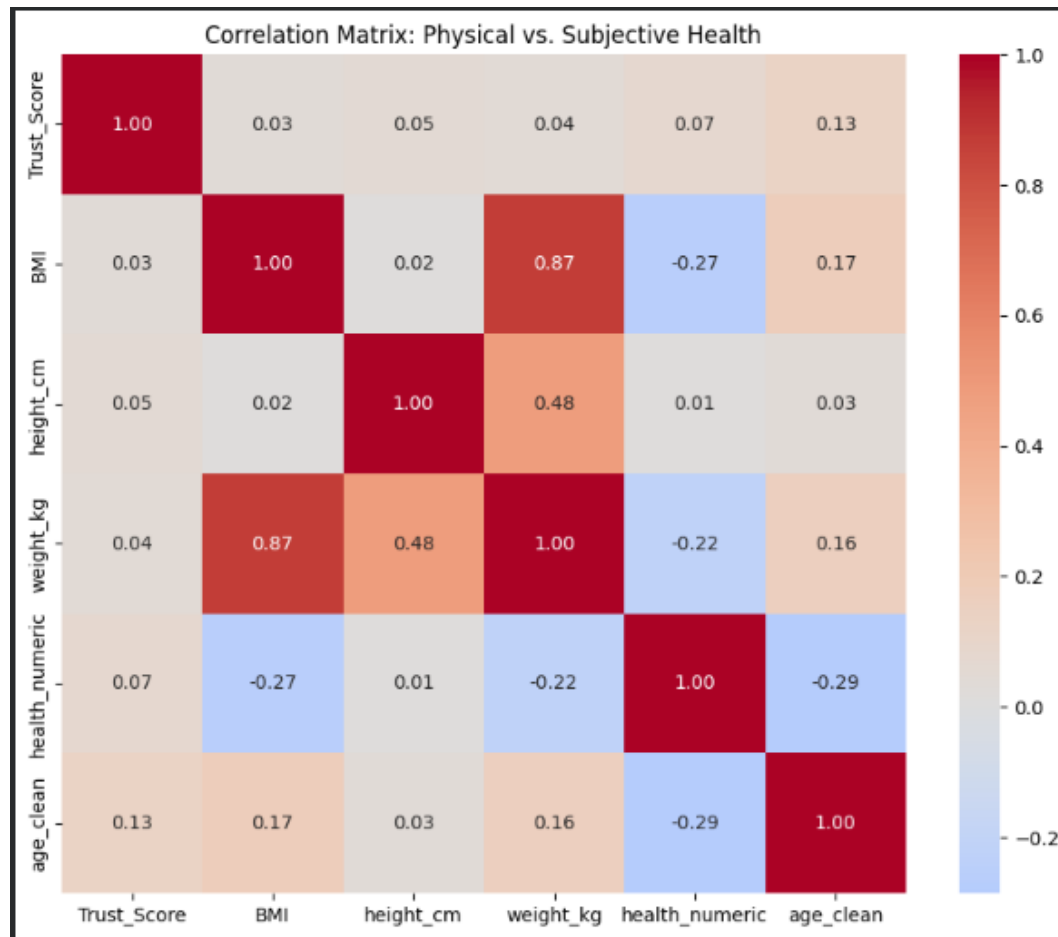
#Cleaning data
df_model = df_isr[existing_features + ['Trust_Score']].copy()
df_model.replace([98, 99, 'Nan', 'nan'], np.nan, inplace=True)

#Filling missing data with median
numeric_cols = ['age_clean', 'height_cm', 'weight_kg', 'BMI']
for col in numeric_cols:
    if col in df_model.columns:
        df_model[col] = df_model[col].fillna(df_model[col].median())

if 'il_rinc' in df_model.columns:
    df_model['il_rinc'] = df_model['il_rinc'].fillna(df_model['il_rinc'].mode()[0])

if 'v47_fixed' in df_model.columns:
    df_model['v47_fixed'] = df_model['v47_fixed'].fillna(0)
```

Correlation Matrix



Conclusions: Health vs. Trust



Hypothesis:

We hypothesized general health would have a higher link to trust.



In Reality:

We could not find a strong relationship between one's health and their view on the system.



Interpretation:

The question of trust is chaotic. We need more data.

The 'Dream Team' Algorithm

- Method: Recursive Feature Elimination (RFE).
- Process:
 - Scanning all variables to mathematically identify the optimal combination.
- Goal:
 - Uncover hidden factors we haven't considered manually.

```
#Find top predictors
for k in range(1, len(features_pool) + 1):

    model = LinearRegression()
    selector = RFE(estimator=model, n_features_to_select=k, step=1)
    selector.fit(X_train, y_train)

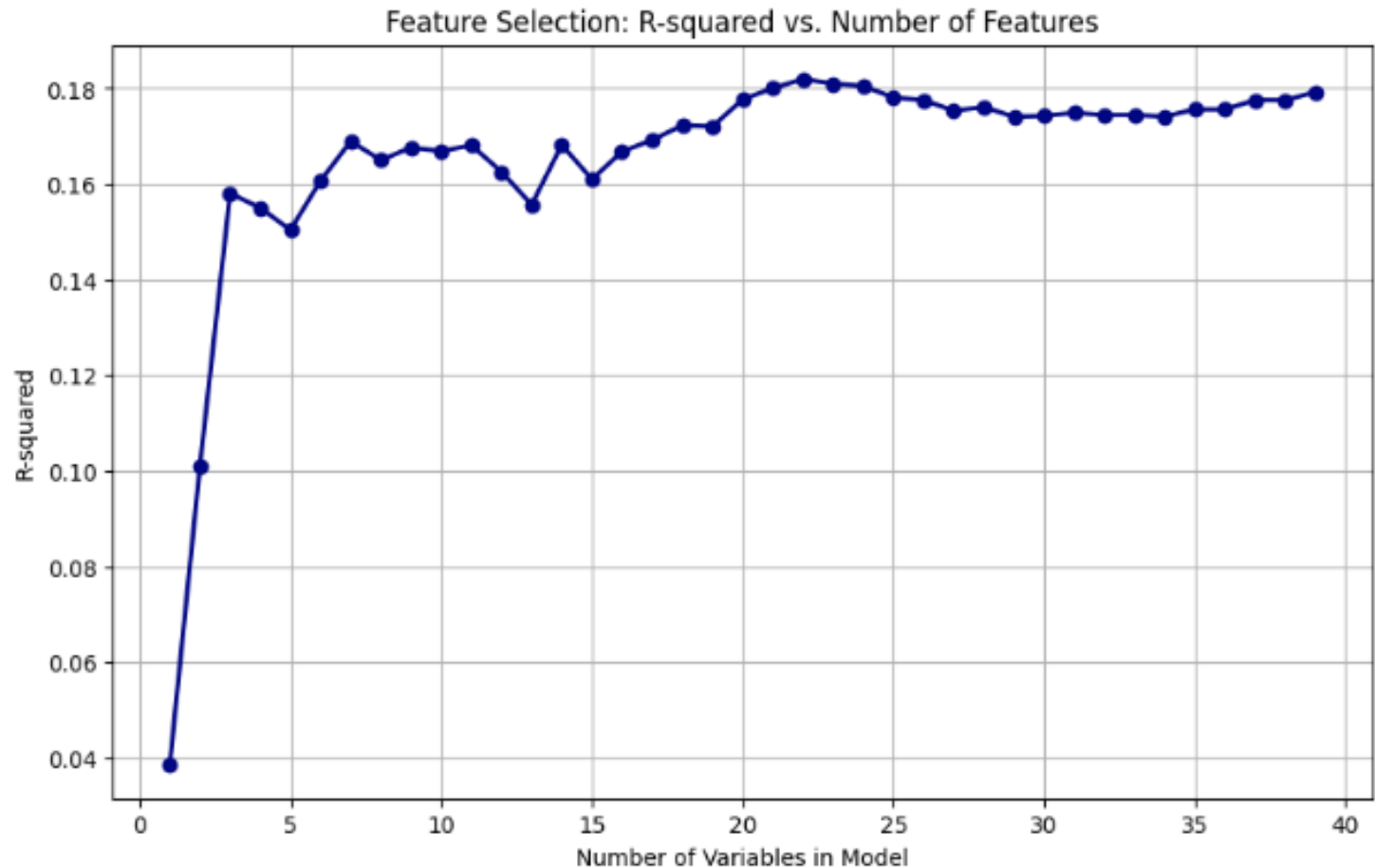
    #Filter chosen predictors
    current_cols = [f for f, s in zip(features_pool, selector.support_) if s]

    #Train on chosen predictors
    model.fit(X_train[current_cols], y_train)
    score = model.score(X_test[current_cols], y_test)

    history.append({'k': k, 'R2': score})

    if score > best_score:
        best_score = score
        best_subset = current_cols
```

Feature Selection Visualization



The Glass Ceiling



Hypothesis:

A purely Data-Driven approach would uncover hidden patterns.



In Reality:

Even with the 'Dream Team' of variables, we hit a ceiling at R-squared ≈ 0.182 .



Interpretation:

The issue isn't data quality, but population heterogeneity.

We are trying to force a 'one-size-fits-all' formula on diverse Israeli society.

Analysis by Demographics

Strategy:

- Break the population into specific sectors to see if the model performs better locally.

Variables Analyzed:

- Gender
- Ethnicity
- Income Groups

```
#If a group is too small (50 people or less) we won't check them.
def check_segment(segment_name, condition):
    subset = df_model[condition]
    if len(subset) < 50:
        return

    X_seg = pd.get_dummies(subset[existing_features], drop_first=True)

    valid_cols = [c for c in best_subset if c in X_seg.columns]

    if not valid_cols: return

    X_seg = X_seg[valid_cols]
    y_seg = subset['Trust_Score']

    X_train_s, X_test_s, y_train_s, y_test_s = train_test_split(X_seg, y_seg, test_size=0.2, random_state=42)

    model.fit(X_train_s, y_train_s)
    score = model.score(X_test_s, y_test_s)

    print(f"Segment: {segment_name:<20} | Count: {len(subset):<4} | R-squared: {score:.3f}")
```

Sector Analysis

Conclusions

- The 'Rational' Sector (Arabs):
 - R-squared ≈ 0.455 . Major breakthrough. Trust is structured and predictable.
- The 'Chaotic' Sectors:
 - Low-Income Groups ($R^2 \approx -1.78$) and Women ($R^2 \approx -0.10$). Model collapsed.
- The 'Ambiguous' Middle:
 - Mizrachi Jews (0.068) and General Jewish (0.073). Model fails to provide meaningful prediction.

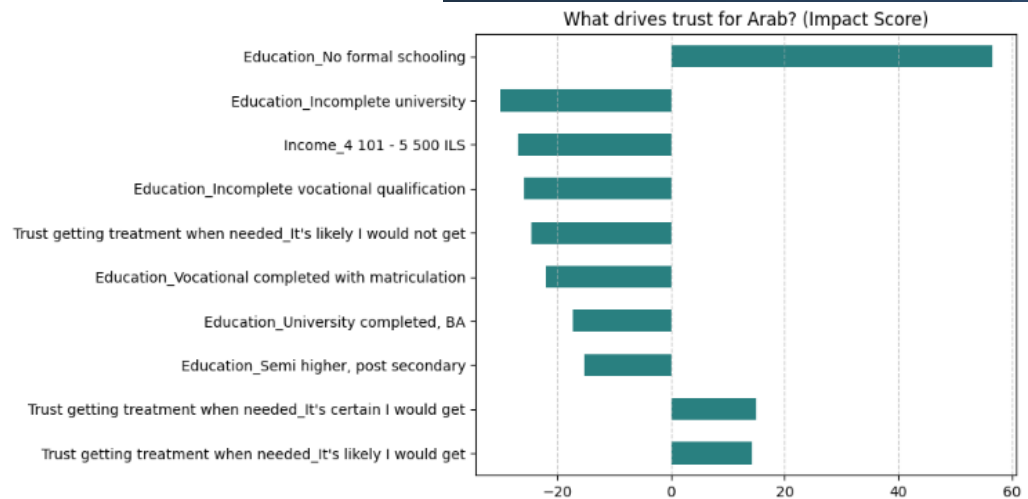
```
--- Analysis by Gender ---
Segment: Sex=Female      | Count: 581 | R-squared: -0.105
Segment: Sex=Male       | Count: 507 | R-squared: 0.141

--- Analysis by Ethnicity ---
Segment: Ethnicity=Ashkenazi | Count: 139 | R-squared: 0.213
Segment: Ethnicity=Other    | Count: 356 | R-squared: -0.043
Segment: Ethnicity=Jewish (General) | Count: 358 | R-squared: 0.073
Segment: Ethnicity=Mizrachi | Count: 142 | R-squared: 0.068
Segment: Ethnicity=Arab     | Count: 93  | R-squared: 0.455

--- Analysis by Income Group ---
Segment: Income=More than 21 | Count: 133 | R-squared: 0.189
Segment: Income=11 701 - 15 1 | Count: 251 | R-squared: 0.056
Segment: Income=2 001 - 4 100 | Count: 80  | R-squared: -0.807
Segment: Income=9 501 - 11 70 | Count: 136 | R-squared: 0.030
Segment: Income=5 501 - 6 700 | Count: 69  | R-squared: 0.033
Segment: Income=15 101 - 21 5 | Count: 109 | R-squared: -0.788
Segment: Income=6 701 - 8 000 | Count: 81  | R-squared: 0.172
Segment: Income=8 001 - 9 500 | Count: 101 | R-squared: 0.142
Segment: Income=4 101 - 5 500 | Count: 56  | R-squared: -1.783
```

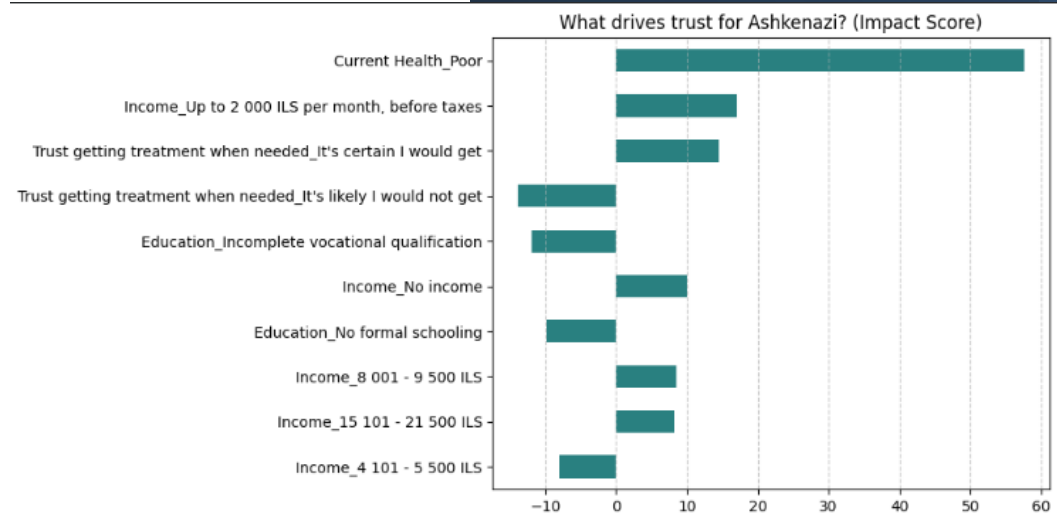
Decoding the Arab Sector

- Theme: 'The Socioeconomic Ladder'
- Key Drivers: Income & Education.
- Insight:
 - Trust is highly rational here.
 - Higher status leads to higher trust.
 - If the system 'works' for them financially, they trust it.



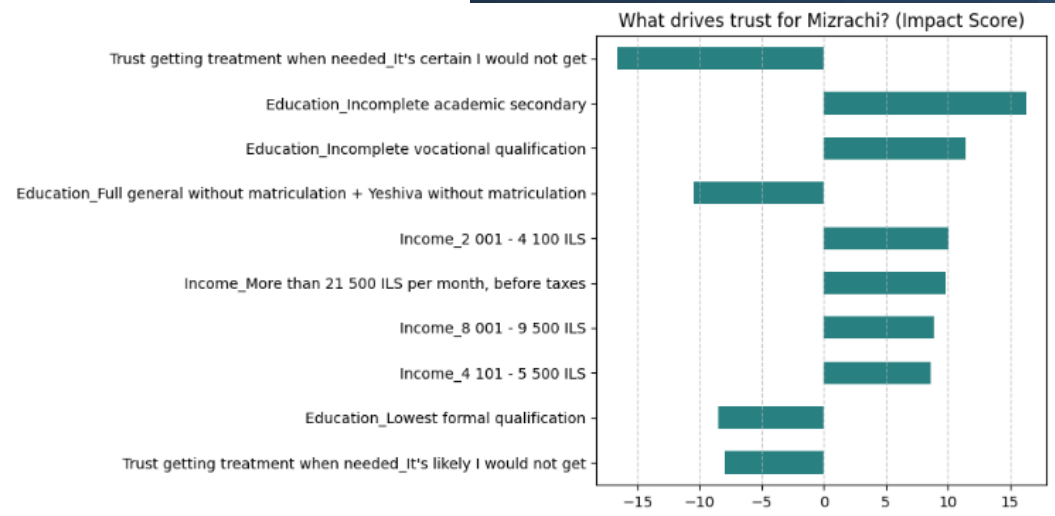
Decoding the Ashkenazi Sector

- Theme: 'The Disappointed Customer'
- Key Driver: Self-Reported 'Poor Health'.
- Insight:
 - Socioeconomic factors matter less.
 - Trust is performance-based.
 - If health declines, trust declines.



Decoding the Mizrachi Sector

- Theme: 'Institutional Skepticism'
- Key Driver: Expectation of treatment.
- Insight:
 - The most sensitive group.
 - Trust is not about money or health results, but Fairness.
 - Strongest predictor of low trust is fear of discrimination.



Final Exploration: Binary Classification

Pivot:

- Predicting exact scores (Regression) struggled with variance.

New Approach:

- Does this person trust the system or not? (Binary)

Classification Split:

- High Trust (1):
Score ≥ 75
- Low/Medium Trust (0):
Score < 75

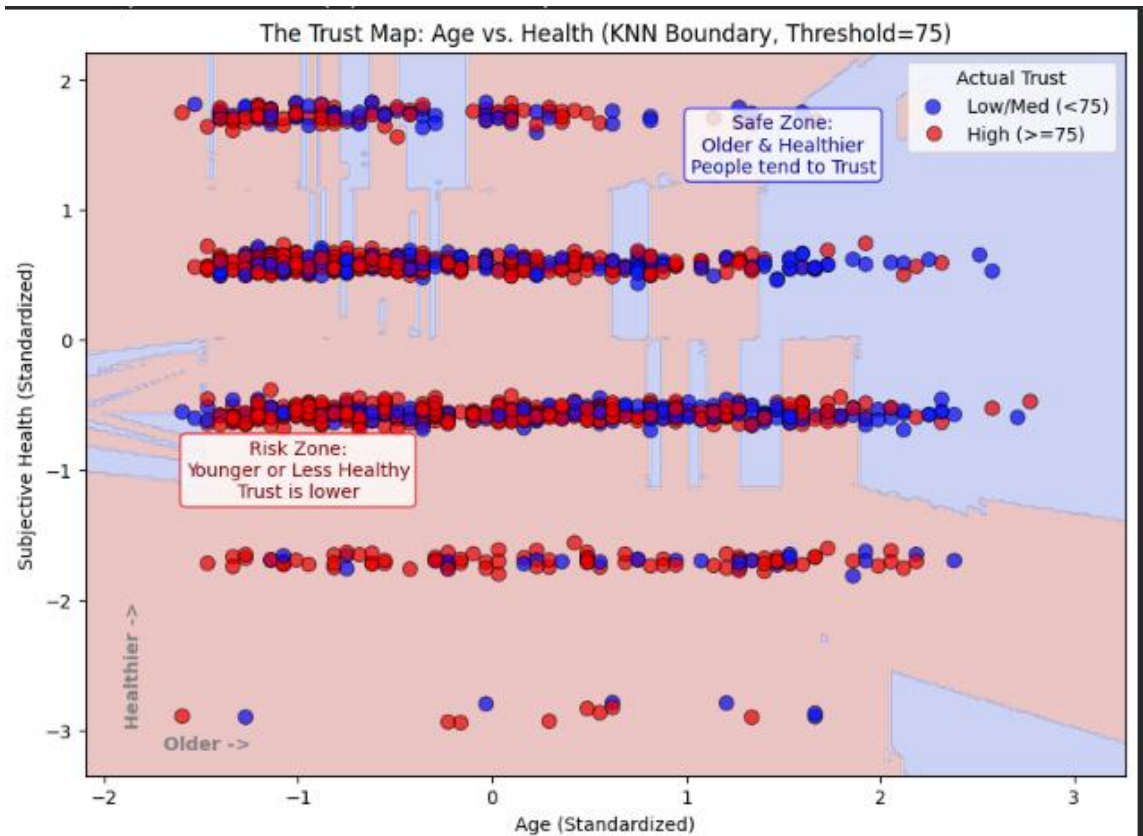
```
#Train KNN !
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_vis)

# K=15 (nearest)
knn = KNeighborsClassifier(n_neighbors=15)
knn.fit(X_scaled, y_vis)
```

KNN Classification Model

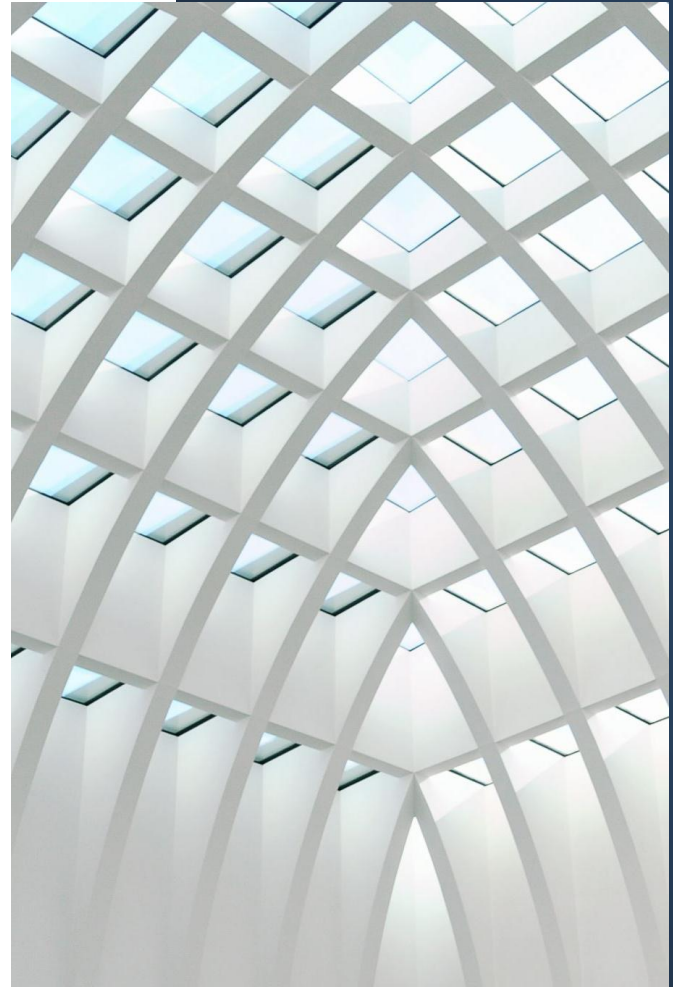
- Model: K-Nearest Neighbors (KNN)
- Features: Age vs. Health
- Goal:
 - Visualize the 'Territory of Trust' by categorizing the population based on age and health status.

The Territory of Trust



Conclusion: The Anatomy of Trust

- The 'Glass Ceiling' is Real:
 - Demographics alone cannot fully explain trust.
- Context is Key:
 - The Arab sector operates on a rational 'Socioeconomic Ladder'.
 - Other sectors are driven by emotional/cultural factors.
- The 'Territory of Trust':
 - Trust is not randomly distributed; it is occupied largely by the older and healthier population.



Final Verdict

- Policy Recommendation:
 - Policymakers cannot rely on general averages.
 - They must address unique community 'pain points':
 - 1. Economic mobility for some.
 - 2. Assurance of fair treatment for others.